1.

(a) $x^4 = x^3 + 10$

$f(x) := x^4 - x^3 - 10 = 0$

$f(2) = -2$

$f(3) = 44$

polynomials always continuous in $\mathbb{R}^n$.

$\Rightarrow$ by IVT, since $-2 < 0 < 44$, $\exists c \in [2,3]$ s.t $f(c) = 0$.

$[2,3]$ is length 1.

(b) Recall thm: absolute error $|x_c - r| \leq \frac{b-a}{2^{n+1}}$

suppose n iterations are needed.

$$\frac{3-2}{2^{n+1}} \leq 10^{-10} \iff \frac{1}{2^{n+1}} \geq 10^{-10}$$

$$\iff 2^{n+1} \geq 10^{10} \iff 2^n \geq 10^{10}/2$$

$$\iff n \geq \log_2(10^{10}) - \log_2 2 \iff n \geq 10\log_2 10 - \log_2(2)$$

$\iff n \geq \frac{10\ln(10)}{\ln(2)} - 1 \approx 32.2193$

$\implies$ need 33 iterations.

2.

(a) $g_1(x) = \frac{1}{2}x + \frac{1}{x}$

$g_1(x_0) = \frac{1}{2} + 1 = \frac{3}{2} = x_1 \approx 1.417$

$g_1(x_1) = \frac{3}{4} + \frac{2}{3} = \frac{17}{12} = x_2 \approx 1.414$

$g_1(x_2) = \frac{17}{24} + \frac{12}{17} = \frac{577}{408} = x_3 \approx 1.4142$

$x_3 = \frac{577}{408} \approx 1.4142$

(b) $g_2(x) = \frac{2}{3}x + \frac{2}{3x} = \frac{2}{3}\left(x + \frac{1}{x}\right)$

$g_2(x_0) = \frac{2}{3}(1+1) = \frac{4}{3} = x_1 \approx 1.333$

$g_2(x_1) = \frac{2}{3}\left(\frac{4}{3} + \frac{3}{4}\right) = \frac{25}{18} = x_2 \approx 1.389$

$g_2(x_2) = \frac{2}{3}\left(\frac{25}{18} + \frac{18}{25}\right) = \frac{949}{675} = x_3 \approx 1.406$

$x_3 = \frac{949}{675} \approx 1.4059$

3.

(a)

(i) $f(x) = x^3 + x^2 - 1$, $f'(x) = 3x^2 + 2x$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$x_0 = 1$

$x_1 = 1 - \frac{1}{5} = 4/5 = 0.8000$

$x_2 = 0.8 - \frac{0.152}{3.52} \approx 0.7568$

(ii) $f(x) = \cos(x^2) - x$, $f'(x) = -2x\sin(x^2) - 1$

$x_0 = 1$

$x_1 = 1 - \frac{\cos(1) - 1}{-2\sin(1) - 1} \approx 0.8287$

$x_2 = 0.8287 - \frac{\cos(0.8287^2) - 0.8287}{-2\sin(0.8287^2) - 1} \approx 0.8043$

(b)

(i) $x_{i+1} = x_i - \dfrac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$

$x_0 = 0$

$x_1 = 1$

$x_2 = 1 - \dfrac{(1^3 + 1^2 - 1)(1-0)}{(1^3 + 1^2 - 1) - (-1)} = \dfrac{1}{2}$

$x_3 = \dfrac{1}{2} - \dfrac{(\frac{1}{2}^3 + \frac{1}{2}^2 - \frac{1}{2})(\frac{1}{2} - 1)}{(\frac{1}{2}^3 + \frac{1}{2}^2 - 1) - (1^3 + 1^2 - 1)} \approx 0.6923$

(ii) $x_0 = 0$

$x_1 = 1$

$x_2 = 1 - \dfrac{(\cos(1) - 1)(1-0)}{(\cos(1) - 1) - (\cos(0) - 0)} \approx 0.6851$

$x_3 = 0.6851 - \dfrac{(\cos(0.6851) - 0.6851)(0.6851 - 1)}{(\cos(0.6851) - 0.6851) - (\cos(1) - 1)}$

$x_3 \approx 0.7828$

4. F contractive $[a,b]$ to $[a,b]$

$\Rightarrow \exists\, 0 \le L < 1$ s.t. $|F(x)-F(y)| \le L|x-y|$

$\forall\, x,y \in [a,b]$

Also, F has unique fixed point s.

$x_{n+1} = F(x_n)$

iterate:

$$|x_{n+1} - x_n| = |F(x_n) - F(x_{n-1})| \le L|x_n - x_{n-1}|$$

repeating gives

$$|x_{n+1} - x_n| \le L^n |x_1 - x_0|$$

now substitute $x_{n+1} = x_n$, $x_n = s$

$$|x_n - s| \le L^n |x_0 - s|$$

$$C := |x_0 - s|, \quad \lambda := L$$

$$|x_n - s| \le \lambda^n C$$

Since $x_0, s \in [a,b]$, $|x_0 - s| \le b-a$

$\Rightarrow C \le b-a$ upper bound for C.



5.

(a) $e_n = r - c_n$

note: $|b_0 - a_0| = b_0 - a_0$

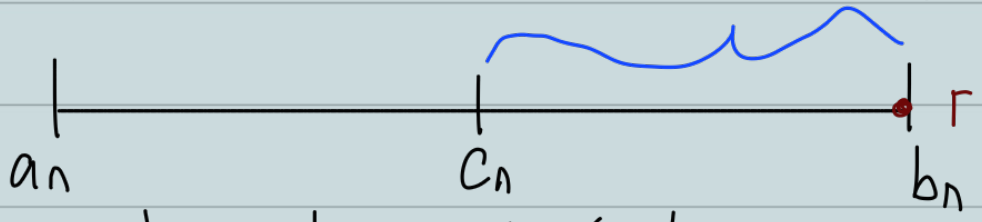
$|b_1 - a_1| = \frac{1}{2}(b_0 - a_0)$

$|b_2 - a_2| = \frac{1}{2}\left(\frac{1}{2}(b_0 - a_0)\right) = \frac{1}{2^2}(b_0 - a_0)$

$\vdots$

$|b_n - a_n| = \frac{1}{2^n}(b_0 - a_0) = \frac{1}{2^{n-1}}(b_1 - a_1)$

since r is inside $[a_n, b_n]$ $\forall n$ and c_n is the midpoint of the interval, $|r - c_n| \leq \frac{b_n - a_n}{2}$ because the max distance c_n can be from r is half the interval $[a_n, b_n]$



$$\Rightarrow |e_n| = |r - c_n| \leq \frac{b_n - a_n}{2}$$

$$= 2^{-(n+1)}(b_0 - a_0)$$

$$= 2^{-n}(b_1 - a_1)$$

$$\Rightarrow |e_n| \leq 2^{-n}(b_1 - a_1)$$

∎

(b) as $n \to \infty$, $c_n \to r$

$\Rightarrow |e_n| = |r - c_n| \to 0$

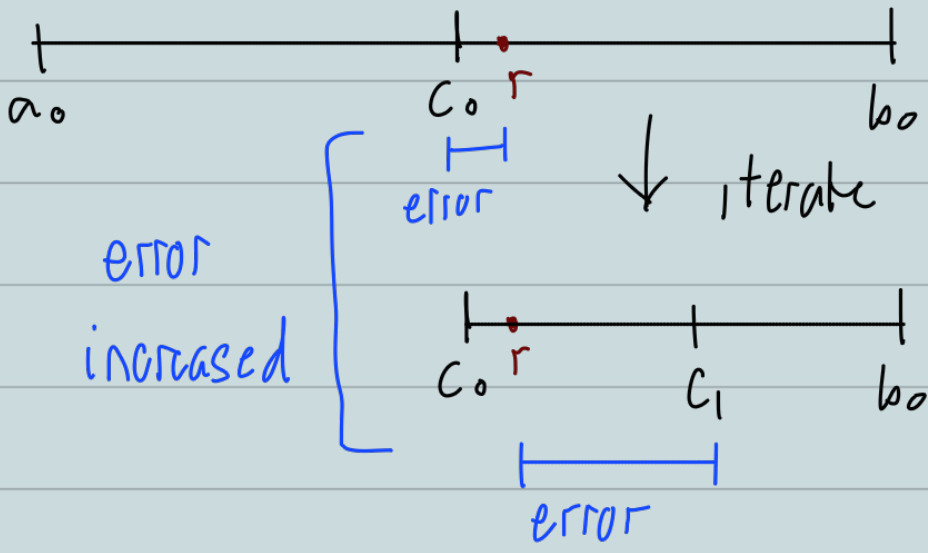
(from Wikipedia definition of Big O notation)

since $b_1 - a_1$ is a constant,

$\exists\, C := b_1 - a_1$ s.t. $|e_n| \leq C 2^{-n} \quad \forall n \in \mathbb{Z}^+$

$\Rightarrow e_n = \mathcal{O}(2^{-n})$

(c) No, since the interval can switch after each iteration, and the new midpoint can be further than the original from r.



$e_n \leq C 2^{-n}$ only guarantees the upper bound to decrease as n increases.

6.

(a) $h(x) := \dfrac{f(x)}{f'(x)}$

$$h(x) = 0 = \frac{f(x)}{f'(x)} = \frac{(x-r)^m p(x)}{m(x-r)^{m-1} p(x) + (x-r)^m p'(x)}$$

$$= \frac{(x-r)\, p(x)}{m p(x) + (x-r) p'(x)}$$

$$\Rightarrow h(x) = (x-r) \frac{p(x)}{m p(x) + (x-r) p'(x)}$$

$\Rightarrow$ r is a simple root of $h(x)$.

(b)

$f(x) = (x-r)^m p(x)$, $p(r) \neq 0$

$f'(x) = m(x-r)^{m-1} p(x) + (x-r)^m p'(x)$

$f''(x) = m(m-1)(x-r)^{m-2} p(x) + m(x-r)^{m-1} p'(x)$
$+ m(x-r)^{m-1} p'(x) + (x-r)^m p''(x)$

$g_{newt}(x) = x - h(x)$

$g'_{newt}(x) = 1 - \dfrac{f'^2 - f f''}{f'^2}$

$= \dfrac{f f''}{f'^2}$

$$g_{newt}'(x) = \frac{[(x-r)^m p(x)]\left[m(m-1)(x-r)^{m-2}p(x) + m(x-r)^{m-1}p'(x) + m(x-r)^{m-1}p'(x) + (x-r)^m p''(x)\right]}{\left[m(x-r)^{m-1}p(x) + (x-r)^m p'(x)\right]^2}$$

$$g_{newt}'(x) = \frac{p(x)\left[m(m-1)(x-r)^{m-2}p(x) + 2m(x-r)^{m-1}p'(x) + (x-r)^m p''(x)\right]}{(x-r)^m\left[m(x-r)^{-1}p(x) + p'(x)\right]^2}$$

$$= \frac{p(x)\left[m(m-1)(x-r)^{-2}p(x) + 2m(x-r)^{-1}p'(x) + p''(x)\right]}{\left[m(x-r)^{-1}p(x) + p'(x)\right]^2}$$

$$= \frac{p(x)\left[m(m-1)p(x) + 2m(x-r)p'(x) + p''(x)(x-r)^2\right]}{(x-r)^2\left(m^2(x-r)^{-2}p(x)^2 + 2m(x-r)^{-1}p(x) + p'^2(x)\right)}$$

$$= \frac{p(x)\left[m(m-1)p(x) + 2m(x-r)p'(x) + p''(x)(x-r)^2\right]}{m^2p(x)^2 + 2m(x-r)p(x) + p'^2(x)(x-r)^2}$$

$$g_{newt}'(r) = \frac{p(r)\,m(m-1)\,p(r)}{m^2\,p(r)^2} = \frac{m-1}{m}$$

$\Rightarrow$ by FPI theorem, local linear convergence with rate $\frac{m-1}{m}$

(c) $g_{mod}(x) = x - mh$

$$g'_{mod}(x) = 1 - m\left(1 - \frac{ff''}{f'^2}\right)$$

$$g'_{mod}(x) = 1 - m\left(1 - g'_{newt}(x)\right)$$

$$g'_{mod}(r) = 1 - m\left(1 - \frac{m-1}{m}\right)$$

$$= 1 - m\left(1 + \frac{1-m}{m}\right)$$

$$= 1 - \left(m + 1 - m\right)$$

$$= 1 - m - 1 + m = 0$$

$\Longrightarrow$ by FPI, local convergence

$e_n = x_n - r$

$$e_{n+1} = x_{n+1} - r = x_n - \frac{m f(x_n)}{f'(x_n)} - r$$

$$= x_n - \frac{m(x_n - r)^m p(x_n)}{m(x_n - r)^{m-1} p(x) + (x_n - r)^m p'(x_n)} - r$$

$$= x_n - \frac{m e_n^m \, p(x_n)}{m e_n^{m-1} p(x_n) + e_n^m p'(x_n)} - r$$

$$= e_n - \frac{m e_n^m \, p(x_n)}{m e_n^{m-1} p(x_n) + e_n^m p'(x_n)}$$

$$e_{n+1} = \frac{e_n^2 p'(x_n)}{m p(x_n) + e_n p'(x_n)}$$

$$\frac{e_{n+1}}{e_n^2} = \frac{p'(x_n)}{m p(x_n) + e_n p'(x_n)}$$

$x_n \to r$ by local convergence

$$\lim_{n \to \infty} \left| \frac{e_{n+1}}{e_n^2} \right| = \frac{p'(r)}{m p(r)}$$

$p(r) \neq 0 \implies$ limit is finite

$\implies$ quadratically convergent.